

BAICHEN YU

Beijing, China

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Education

Peking University

Ph.D. Candidate in Statistics (Applied Economics) at Guanghua School of Management

09 2023–
Beijing, China

Advisor: Prof. Hansheng Wang

The Hong Kong University of Science and Technology

Visiting Research Postgraduate Student at Department of Mathematics

07 2025– 12 2025
Hong Kong SAR, China

Advisor: Prof. Can Yang

East China Normal University

B.Sc. in Statistics - GPA: 3.84/4 - Rank: 1/86

09 2019 – 06 2023
Shanghai, China

Research Interests

- Methodology: Spatial statistics; High-dimensional factor models; Multiple instance learning
- Application: Image data analysis; Network data analysis; Biomedical data analysis; Financial technology

Research and Publications

PEER-REVIEWED JOURNAL ARTICLES

A Semiparametric Gaussian Mixture Model with Spatial Dependence and Its Application to Whole-Slide Image Clustering Analysis

Baichen Yu, Jin Liu, and Hansheng Wang*

Biometrics, 2025, Volume 81, Issue 4: uja149.

A Geometrical Model for Abnormal Motion Detection of Portal Crane Bucket Grab

Baichen Yu, Xiao Wang and Hansheng Wang*

Engineering Applications of Artificial Intelligence, 2025, Volume 139, Part A: 109481.

Detecting Breast Carcinoma Metastasis on Whole-Slide Images by Partially Subsampled Multiple Instance Learning

Baichen Yu, Xuotong Li, Jing Zhou and Hansheng Wang*

The Annals of Applied Statistics, Accepted.

High-Dimensional Spatial Autoregression with Latent Factors by Diversified Projections

Jiaxin Shi, Xuening Zhu, Jing Zhou, Baichen Yu and Hansheng Wang*

Journal of the American Statistical Association, 2026, Volume 121, Issue 554: 1232–1243.

PREPRINTS AND WORKING PAPERS

Characterizing dynamic tissue architectures by identifying cell-type-specific spatiotemporal gene programs with stGP

*Baichen Yu, Ziyue Tan, Xiaomeng Wan, Hansheng Wang, and Can Yang**

Under review at Nature Communications

Variational inference methods for single-cell genomics

*Baichen Yu[#], Ziyue Tan[#], Hu Che, and Can Yang**

Under review at Statistical Learning and Data Science

A unified framework for integrating single-cell and spatial transcriptomics to uncover spatiotemporal tissue architecture

*Ziyue Tan[#], Baichen Yu[#], and Can Yang**

In preparation

Conference and Invited Talks

- ICSA China Conference, 2023 and 2024
- The 7th International Conference on Econometrics and Statistics (EcoSta 2024)
- Hangzhou International Conference on Frontiers of Data Science, 2023
- The 2nd Joint Conference on Statistics and Data Science in China, 2024
- The 17th China-R Conference, 2024
- The 25th and 26th Beijing-Tianjin-Hebei Youth Seminar on Probability and Statistics, 2023 and 2025
- The 6th and 7th North China Statistics and Optimization Conference, 2024 and 2025
- The 2nd, 3rd, 4th Annual Meeting of Chinese Statistical Association of Young Scholars, 2024, 2025, and 2026
- Online seminar talk at Clubear (8,000+ views), 2024, 2025, and 2026

Teaching Experience

Teaching Assistant for

- Regression Analysis (UG) - 2022 Fall@ECNU
- Deep Learning and Artificial Intelligence (UG & PG & MBA) - 2023, 2024 Fall@PKU
- Business Data Analysis and Forecasting (MBA) - 2024 Spring@PKU
- Academic Writing (BA) - 2024, 2025, 2026 Spring@PKU
- Python Fundamentals for Data Science (UG & PG) - 2024 Fall@PKU

Honors

- Merit Student of Guanghua School of Management, Peking University
- Outstanding Teaching Assistant Award, Guanghua School of Management, Peking University
- Shanghai Outstanding Graduates (Summa Cum Laude)
- First-class Scholarship of East China Normal University

Service

Reviewer for

- *Statistica Sinica*
- *Engineering Applications of Artificial Intelligence*
- *Advanced Engineering Informatics*

Skills

- **Programming Languages:** R, Python (TensorFlow & PyTorch), C++, Julia, SQL, $\text{\LaTeX}$
- **Natural Languages:** Chinese, English